

Props and Rocks: Passive Haptic Mixed Reality for Navigating Far-off Worlds

Michael Vincenty
mdv29@njit.edu

New Jersey Institute of Technology
Newark, New Jersey, USA

Joshua Grebler
jhg29@njit.edu

New Jersey Institute of Technology
Newark, New Jersey, USA

Cindy Piza
cnp3@njit.edu

New Jersey Institute of Technology
Newark, New Jersey, USA

Ozara Dalgo
omd4@njit.edu

New Jersey Institute of Technology
Newark, New Jersey, USA

Taro Narahara
narahara@njit.edu

New Jersey Institute of Technology
Newark, New Jersey, USA

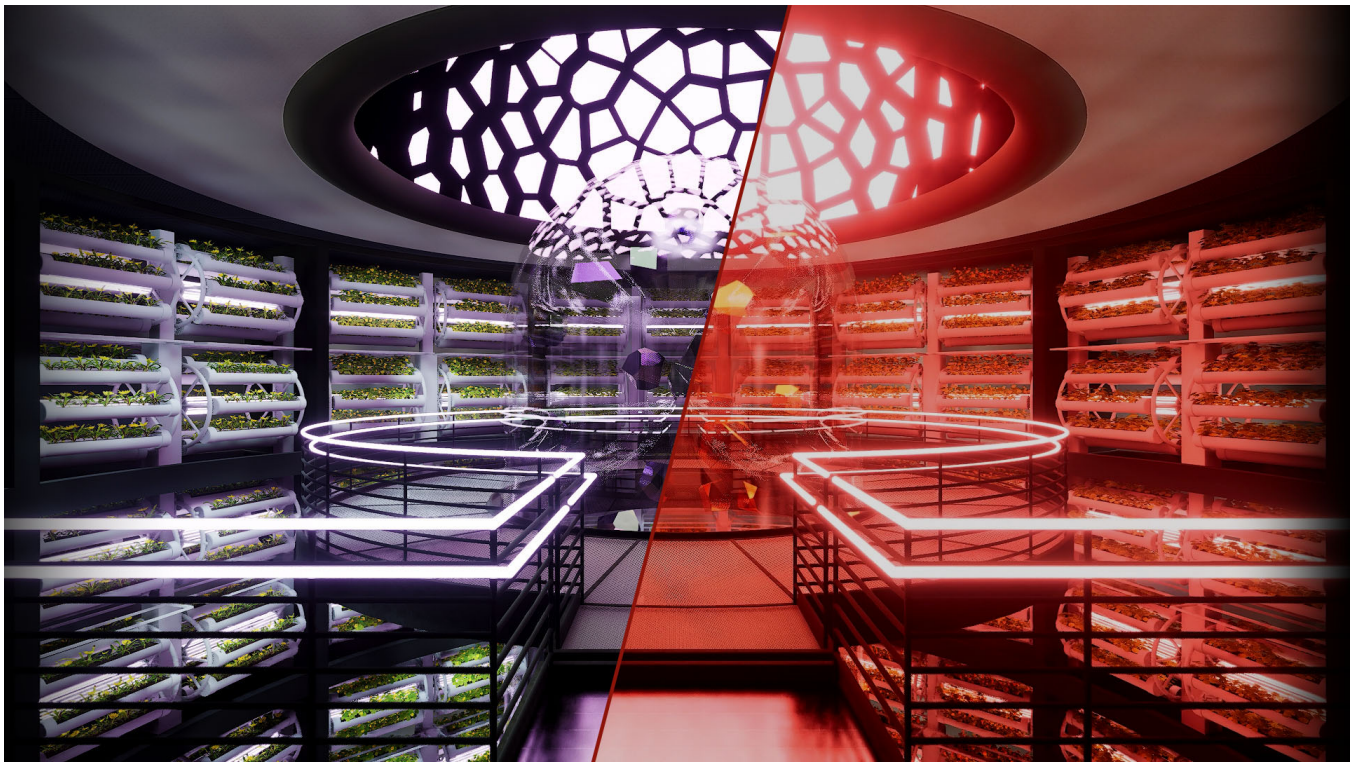


Figure 1: Player's perspective view displaying the vertical hydroponic farm in need of restoration.

ABSTRACT

Our project immerses participants in a world blending virtual and physical realities. Users navigate a divided society, interacting with props via haptic feedback. Employing innovative techniques, including redirected walking, our project seamlessly transitions between virtual worlds, advancing VR integration.

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).

SIGGRAPH Immersive Pavilion '24, July 27–August 01, 2024, Denver, CO, USA

© 2024 Copyright held by the owner/author(s).

ACM ISBN 979-8-4007-0527-4/24/07

<https://doi.org/10.1145/3641521.3664404>

CCS CONCEPTS

• **Human-centered computing** → **Virtual reality**; *Haptic devices*.

KEYWORDS

virtual reality, haptic feedback, redirected walk

ACM Reference Format:

Michael Vincenty, Joshua Grebler, Cindy Piza, Ozara Dalgo, and Taro Narahara. 2024. Props and Rocks: Passive Haptic Mixed Reality for Navigating Far-off Worlds. In *Special Interest Group on Computer Graphics and Interactive Techniques Conference Immersive Pavilion (SIGGRAPH Immersive Pavilion '24)*, July 27–August 01, 2024, Denver, CO, USA. ACM, New York, NY, USA, 2 pages. <https://doi.org/10.1145/3641521.3664404>



Figure 2: Series of spaces showing progressions from the underworld to above. Redirected walk technique with haptics and non-Euclidean space beyond the window deceiving players' perception.

1 INTRODUCTION

Faced with the challenge of envisioning future environments for the next generation, our team crafted a vision of a society thriving off-world. Our concept centers around a luxury city settlement in a distant world, the affluent City of *Allurali*, providing a canvas to explore humanity's potential future. However, this city has its secrets. Our narrative mirrors humanity's cyclical over-exploitation of the environment, a critical contemporary concern. We envisioned a scenario where Earth's dwindling resources have incentivized humans to move off the world and chose to convey our narrative through an interactive VR experience. Set many years after the initial colonization of the planet, our world is divided into two distinct classes: the privileged "*Elites*" residing in an above-ground sanctuary shielded from the hazardous external environment, and the exiled "*Undesirables*" condemned to toil in subterranean caves, clad in environmental hazard suits mandated by the government's rating system. Many elite citizens opt to recreate semblances of their former lives on Earth, using advanced technology to simulate lush landscapes outside their windows and doors, blurring the lines between the metaverse and reality, thereby causing confusion and distorting perceptions of physical space.

At a pivotal moment, a malfunction in the AI-controlled air purification system, which had isolated the city from the outside world, led to the demise of the *Elites*. This catastrophic event left only the lower-class citizens laboring underground in protective gear, and the remnants of the upper world—a merging of virtual and real spaces—abandoned by the *Elites*. In our immersive VR project, participants assume the roles of the remaining "*Undesirables*," embarking on a journey from the depths of the underground to the upper realm, striving to restore order amid chaos. Their mission: repair the malfunctioning AI and rejuvenate a colossal vertical hydroponic farm, the lifeline of the city's inhabitants.

2 DESIGN AND IMPLEMENTATION

Players can only discern between the increasingly distorted virtual world, a product of the AI malfunction left behind by the *Elites*, and the physical reality solely by physically interacting with props in VR. Outside windows and doors, expansive landscapes and vistas often deceptively conceal narrow passages leading to the next room, challenging players with spatial puzzles as they advance toward the upper realm. To realize this world, we employed Stencil Buffer and Redirected Walk. To create portals hidden inside doors and windows, we utilized a stencil buffer technique. This technique, assigning stencil buffers to portals, enables seamless transitions between worlds. Stencil buffer, a computer graphics method reducing

GPU and CPU load by omitting the display of unnecessary parts like those hidden in shadows, is commonly used for portals in VR applications [Joyce and Robinson 2017]. Multiple "scenes" or virtual environments are loaded simultaneously, with scenes farther away unloaded as players pass through portals, while upcoming scenes are loaded in, occluded by opacity masks on scene objects' materials. Material-based rendering with opacity masks and stencil buffer functionality limits user visibility to the current scene, allowing us to switch world content based on player interaction. Our virtual scenes mirror a physical environment constructed within the play space, where virtual walls correspond to real-world counterparts for tactile interaction. This mapping facilitates passive haptic interaction, eliminating the need for user-worn haptic hardware, which historically has limitations compared to passive haptic techniques [Langbehn et al. 2018].

Redirected walking [Razzaque 2005] enhances design flexibility in virtual environments while maintaining correspondence with the physical world. Imperceptibly distorting the player's perspective as they navigate alters their path, guiding them toward physical props corresponding to virtual scene elements. For instance, approaching a virtual round table in one scene, not present in the physical set, triggers a haptic experience by aligning the table's edge with a point on a straight desk. Injecting a small rotational gain in locomotion while in proximity to the table creates the illusion of circling it, with the edge consistently corresponding to the physical desk.

3 CONCLUSION

The project began as an experiment within a design studio environment, involving students from various disciplines such as interior design, digital design, industrial design, and information technology, each contributing their expertise. Another notable aspect of the project is the development of modules designed that could be readily applied to any other student's creative projects. These modules focus on portal systems employing stencil buffers and redirected walking to meet the spatial demands integral to today's VR experiences.

REFERENCES

- Richard D Joyce and Stephen Robinson. 2017. Passive haptics to enhance virtual reality simulations. In *ALAA Modeling and Simulation Technologies Conference*. 1313.
- Eike Langbehn, Frank Steinicke, Markus Lappe, Gregory F Welch, and Gerd Bruder. 2018. In the blink of an eye: leveraging blink-induced suppression for imperceptible position and orientation redirection in virtual reality. *ACM Transactions on Graphics (TOG)* 37, 4 (2018), 1–11.
- Sharif Razzaque. 2005. *Redirected walking*. The University of North Carolina at Chapel Hill.